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**Computational Thinking for Problem Solving**

**Handout - Lab Session - 1**

**Exercise 1:** Create a flowchart that has a start and end symbol. Add an "Output" symbol with the text "Hello, World!" connecting to the end symbol.

**Exercise 2:** Create a flowchart that displays your favorite quote on the screen.  
"The only way to do great work is to love what you do." - Steve Jobs

**Exercise 3:** Create a flowchart that displays an art shape, such as a heart, star or a pyramid, using ASCII characters.

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**Exercise 4:** Create a flowchart that prompts the user to enter their age and then displays a message about their age.

**Sample input :** Please Enter your age : *[user enters 18]*

**Sample output :** You are 18 years old.

**Exercise 5:** Create a flowchart that prompts the user for their favorite color and then displays a message about their choice

**Sample input :** Please Enter your favorite color: *[user enters Red]*

**Sample output :** Red is a nice color !

**Exercise 6:** Create a flowchart that prompts the user to enter their birth year and calculates and displays their current age.

**Sample input :** Please Enter your birth year: *[user enters 2006]*

**Sample output :** You are 17 years old !

**Exercise 7:** Create a flowchart that asks the user for their weight (in kilograms) and height (in meters), and then calculates and displays their BMI (Body Mass Index).

**Sample input :** Please enter your weight (in kilograms): *[user enters 70]*

                    Please enter your height (in meters): *[user enters 1.75]*

**Sample output :** Your BMI is 22.86

**Exercise 8:** Create a flowchart that prompts the user to enter two numbers and display the sum of two numbers.

**Sample input :** Please enter first number : *[user enters 20]*

                    Please enter second number : *[user enters 25]*

**Sample output :** The sum is 45.

**Exercise 9:** Create a flowchart that prompts the user to enter a number of seconds and calculates and displays the equivalent time in minutes and seconds.

**Sample input :** Please enter number of seconds : *[user enters 150]*

**Sample output :** 150 seconds is equal to 2 minutes and 30 seconds.

**Exercise 10:** Create a flowchart that converts the distance from miles to kilometers.

**Sample input :** Please enter distance in miles : *[user enters 10]*

**Sample output :** 10 miles is approximately equal to 16.0934 kilometers.